recovery under degraded communications, contested EM environments, or partial system failures.

These principles are formalized in mission logic: operators specify intent and constraints; the onboard autonomy stack translates intent into mission plans, continuously re-optimizes under evolving conditions, and escalates only those decisions that surpass predefined human-in-the-loop thresholds (human-supervised autonomy/human–machine teaming).

5.3.2 Hardware components

Fourth-generation platforms combine mature aerodynamic and propulsion design with advanced sensing, computing, and hardened communications. Key hardware elements include:

- *Airframe and mission modularity*: Low-observable or low-signature airframes for survivability, with modular bays to swap sensors, EW packages, or payloads quickly.
- *Propulsion and power systems*: Efficient hybrid-electric/turboprop or turbofan cores for extended endurance; integrated power management to supply high-power edge compute, directed energy payloads, or extended sensor suites; thermal management for electronics and sensors.
- *Flight control units (FCU) and redundant avionics*: Multi-lane flight computers, hardware-in-the-loop protected buses, and multiple sensor feeds to permit cross-checking and safe degradation.
- *High-performance edge compute*: Ruggedized CPUs/GPUs/TPUs for onboard ML inference, real-time sensor fusion, and simulation-assisted planning.